

ECON 300 – Labour Economics

Quiz 4

March 31, 2026

1. In the Oaxaca decomposition, the portion of the wage gap that is attributed to differences in observed characteristics such as schooling and experience is called:
 - A) the discrimination component
 - B) the unexplained component
 - C) the explained component
 - D) the crowding component
 - E) the coefficient bias component
2. Suppose women and men have the same observed productivity characteristics, but women still receive lower wages because firms statistically expect greater labour force interruptions among women. This is best described as:
 - A) equalizing differences
 - B) taste-based discrimination only
 - C) statistical discrimination
 - D) human capital investment
 - E) comparable worth
3. According to the crowding hypothesis, if women are restricted to a narrow set of occupations, then:
 - A) wages in those occupations will tend to rise because of specialization
 - B) wages in those occupations will tend to fall because labour supply is concentrated there
 - C) men's wages in all occupations will necessarily fall
 - D) occupational segregation has no effect on wages
 - E) wage discrimination disappears in competitive equilibrium

4. Which of the following would most likely reduce the unexplained portion of the gender wage gap in an Oaxaca decomposition?
 - A) adding a variable for years of schooling when schooling differs across groups
 - B) removing occupation controls from the regression
 - C) increasing the sample size without changing the model
 - D) estimating the model without experience
 - E) using nominal instead of real wages

5. Which of the following best describes the **unexplained** component in an Oaxaca decomposition?
 - A) the portion of the wage gap due to differences in education and experience
 - B) the portion of the wage gap due to differences in labour force participation rates
 - C) the portion of the wage gap due to differences in observed characteristics only
 - D) the portion of the wage gap due to differences in coefficients or returns to characteristics
 - E) the portion of the wage gap explained entirely by occupational choice

Problem

Consider the following estimated log annual earnings equations for men and women:

$$\ln W_m = 1.20 + 0.09S + 0.05EXP$$

$$\ln W_f = 0.95 + 0.07S + 0.04EXP$$

where:

- W_m = annual earnings of men
- W_f = annual earnings of women
- S = years of schooling
- EXP = years of labour market experience

Suppose the average characteristics of men and women are:

$$\bar{S}_m = 16, \quad E\bar{X}P_m = 12$$

$$\bar{S}_f = 15, \quad E\bar{X}P_f = 10$$

Answer the following:

- (a) Compute the predicted log earnings for men using the male wage equation. [3 points]
- (b) Compute the predicted log earnings for women using the female wage equation. [3 points]
- (c) Compute the raw gender gap in predicted log earnings:

$$\ln W_m - \ln W_f$$

[2 points]

- (d) Using the **male wage structure** as the reference group, compute the portion of the gap explained by differences in characteristics:

$$(\bar{X}_m - \bar{X}_f)\hat{\beta}_m$$

[4 points]

- (e) Compute the unexplained portion of the gap. [2 points]
- (f) Interpret the explained and unexplained components economically. [1 point]